MALITH SHEHAN SANKALPA

Data Science Undergraduate | Machine Learning & Analytics
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PROFESSIONAL SUMMARY

Fourth year Data Science undergraduate with hands-on experience across the full ML lifecycle, data cleaning, feature engineering, model development, and evaluation spanning healthcare classification, graph neural networks, and time series forecasting. Proficient in Python, SQL, and the Scikit-learn/PyTorch ecosystem, with practical MLOps exposure (experiment tracking, reproducible pipelines) and applied deep learning coursework in CNNs, GNNs, and NLP. Currently leading the AI/data science layer of a 5-member Final Year Project building an AI-powered personal finance application. Seeking a Data Science, Machine Learning, or Data Analyst role to apply predictive modeling and explainable AI skills in an applied setting.

TECHNICAL SKILLS

Programming - Python, SQL, Java (basic), Dart

Machine Learning & Deep Learning - Scikit-learn, XGBoost, PyTorch, Keras/TensorFlow, Graph Neural Networks (GCN, GAT), LSTM, BERT/Transformers, supervised & unsupervised learning, model evaluation & hyperparameter tuning, explainable AI (SHAP, LIME)

Data Analysis & Visualization - Pandas, NumPy, Exploratory Data Analysis, Feature Engineering, Matplotlib, Seaborn, Power BI, Excel

MLOps & Experimentation - DagsHub (experiment tracking & versioning), reproducible pipelines, GitHub Actions, Streamlit

Databases - MySQL, PostgreSQL (basic), Supabase

Cloud / DevOps - Docker (basic), Git, GitHub

Soft Skills - Leadership, technical communication, problem-solving, adaptability, critical thinking

PROJECTS

AI-Based Personalized Financial Coaching & Sandbox Investment Training App (Final Year Project, Ongoing) *Flutter · Dart · Supabase · Python · PyTorch · LSTM · BERT · SHAP/LIME*

- Own the AI and data science layer of a 5-member Final Year Project, supervised by Dr. Chameera De Silva and Mr. Nethum Dilchitha, within a three-layer architecture (Flutter/Dart presentation, decoupled Python AI microservices, Supabase data layer).
- Building an LSTM regression model for next-month expenditure prediction and fine-tuning a BERT model for financial news sentiment classification.
- Applying SHAP/LIME explainability across all model recommendations to support transparent, trustworthy AI, and designing evaluation via SUS scores, MAE, and paired t-tests.

SmartCare Hospital — Clinical Risk Prediction System *Python · Scikit-learn · Logistic Regression · Model Evaluation*

- Developed and evaluated multiple classification models for hospital patient risk prediction as part of a group AI coursework project, covering data preprocessing, model development, and comparative evaluation.
- Selected Logistic Regression as the champion model based on simplicity, interpretability, and native probability calibration, over more complex alternatives.
- Investigated and documented near-perfect model performance to distinguish genuine signal from dataset artifacts, strengthening the model evaluation methodology and final report.

Node Classification on OGBN-Arxiv with Graph Neural Networks *Python · PyTorch Geometric · GCN · GAT · Streamlit*

- Built a full graph neural network pipeline for node classification on the OGBN-Arxiv citation network, implementing and comparing GCN and GAT architectures as part of a group Tensors & Graphs coursework project.
- Developed an interactive Streamlit dashboard to visualize model predictions and performance, alongside a technical report and public GitHub documentation.

End-to-End Customer Churn Prediction with MLOps Pipeline *Python · Pandas · NumPy · Scikit-learn · XGBoost · DagsHub*

- Designed a full ML workflow from raw customer data to a deployable churn prediction model, including ingestion, cleaning, preprocessing, and EDA to identify statistically significant churn indicators.
- Engineered predictive features and applied feature selection to improve model performance, then trained and compared multiple classification models (Scikit-learn, XGBoost) to select the best performer.
- Implemented experiment tracking and model versioning with DagsHub, packaging the workflow into a reproducible, production-ready pipeline.

SAPRO v1.0 — Smart Greenhouse Management System *Backend Development · Real-Time Monitoring · Automation*

- Designed and implemented the backend for a smart greenhouse system enabling real-time environmental monitoring and automated control for smart agriculture.
- Built services to collect, process, and store environmental data, structuring the database for historical tracking and trend analysis, with automation logic to trigger control actions based on threshold breaches.

EDUCATION

BSc (Hons) in Data Science - Sri Lanka Technology Campus (SLTC) *(2024 – 2027)*

Diploma in Information Technology - ESOFT Metro Campus *(2022 – 2023)*

Diploma in English - ESOFT Metro Campus *(2022 – 2023)*

G.C.E. Advanced Level – Physical Science - *(2022)*

CERTIFICATIONS

Machine Learning Specialization - DeepLearning.AI (2026)

Supervised Machine Learning • Advanced Learning Algorithms • Unsupervised Learning, Recommenders & Reinforcement Learning

Cisco Certified Network Associate (CCNA) - Cisco Networking Academy (2026)

Introduction to Networks • Switching, Routing & Wireless Essentials • Enterprise Networking, Security & Automation

Generative AI for Everyone - DeepLearning.AI (2025)

Credential ID: ZD3G9ZSXAX6K